DRONACHARYA COLLEGE OF ENGINEERING, GURUGRAM
PRE-UNIVERSITY EXAMINATION APRIL, 2025
(Gurugram University)

Semester: IV Branch: CSE/CSIT/CSE(AI&ML)

QUESTION PAPER

Subject with Code: R-PROGRAMMING

Time: 3 hours Max. Marks: 70

Note: *Q.1 is compulsory. Attempt any ONE question from each Unit. Each question carries of 14 marks. Assume data, if any missing.*

Q. 1. Objective type / Short-answer type questions. (2x7=14)

i. Why R programming language? What are the applications of the R programming language? [CO1]

ii. Describe installed.package() and find.package() in R programming? [CO1]

iii. Describe the usage of scan( ) and readline ( ) functions in R? [CO3]

iv. Differentiate NA, NaN and NULL. [CO2]

v. Describe the string and also write about substr() and grep() string functions. [CO3]

vi. Explain the factors and the generating factor levels gl() function? [CO4]

vii. Explain the getwd() and setwd() function in R?What do you understand by Data Visualization? What is the ggplot2 package in R? [CO5]

UNIT-I

Q. 2.a) Write an R program to demonstrate the different types of operators in R. [CO2]

(b) How do you assign a variable in R programming? Write the rules for naming a variable. [CO2]

Q. 3. (a) Explain the data types in R programming, with examples. [CO2]

(b) WAP to determine whether a triangle is equilateral, isosceles, or scalene. [CO2]

UNIT-II

Q. 4. (a) Explain the while loop in R programming. Write about the next and break in R programming?[CO3]

(b) Write short notes on the following built-in functions in R programming: - mean(), paste(), sum(), min(), max(), seq(), rep(). [CO3]

Q.5. (a) What is a function? WAP to find the factorial of a number by calling the function without an argument value. [CO3]

(b) Differentiate between list and vector data types in R. Explain the various element wise operations and important functions that can be performed on vectors in R. [CO3]

UNIT-III

Q. 6. a) Discuss in detail about matrices like creating matrices, accessing the matrix elements. Discuss the rbind() and cbind() functions in R. [CO4]

(b) How do you create and merge a data frame in R? Also, discuss the functions: dim(), nrow(), ncol(), str(), Summary(), names(), head(), tail() apply on the data frame. [CO4]

Q.7 (a) Describe the array attributes with an example. How to access and manipulate the array elements. [CO4]

(b) Write about the dataframe in R. Create a data frame named Employee with the columns (Emp_Id, EmpName, Score). Find the mean score of all the employees. Extract the rows for employees who scored more than 75. Add a new column performance Grade based on the Score (e.g. A for score > 80, B for 60-80, C for <60). [CO4]

UNIT-IV

Q. 8 (a) Define plot () function. How to set the axis labels and title of the R plots? [CO6]

(b) How to read and analyze the CSV file data using the functions: summary(), min(), max(), range(), mean(), median(), apply(). [CO5]

Q. 9 (a) How to read and write in a particular file from the working directory. [CO5]

(b) Write about the bar chart, histogram and scatter plot? Describe the components of a box plot. Construct a box plot for the following data with R code: 100, 95, 93, 100, 92, 95, 99, 100, 58, 75, 78, 45, 66, 89, 93. Construct the Box Plot components for the above data. [CO6]